

Creating Hyperlinks

Connecting the Web... One link at a time.

Hyperlinks Using CSS

Plain hyperlink: [Read my Blog](#) - Border added to links to show the size of the "hot" area.

Remove underline + add padding for larger target area: [Read my Blog](#)

Change link color: (all four states: new - visited - hover - active) [Read my Blog](#) [Visit my Site](#)

-- Depends on browser implementation, especially for visited links.

Button Effect: [Read my Blog](#)

Hover Effect: [Read my Blog](#)

Button Effect using a graphic with text: [Like This](#)

Whatever effect you use, be consistent and use the same effect for all the hyperlinks on the page.

Hyperlink Preview

History of the Hyperlink

The term "hyperlink" was coined in 1965 (or possibly 1964) by [Ted Nelson](#) at the start of [Project Xanadu](#). Nelson had been inspired by "[As We May Think](#)", a popular 1945 essay by [Vannevar Bush](#). In the essay, Bush described a microfilm-based machine (the [Memex](#)) in which one could link any two pages of information into a "trail" of related information, and then scroll back and forth among pages in a trail as if they were on a single microfilm reel.

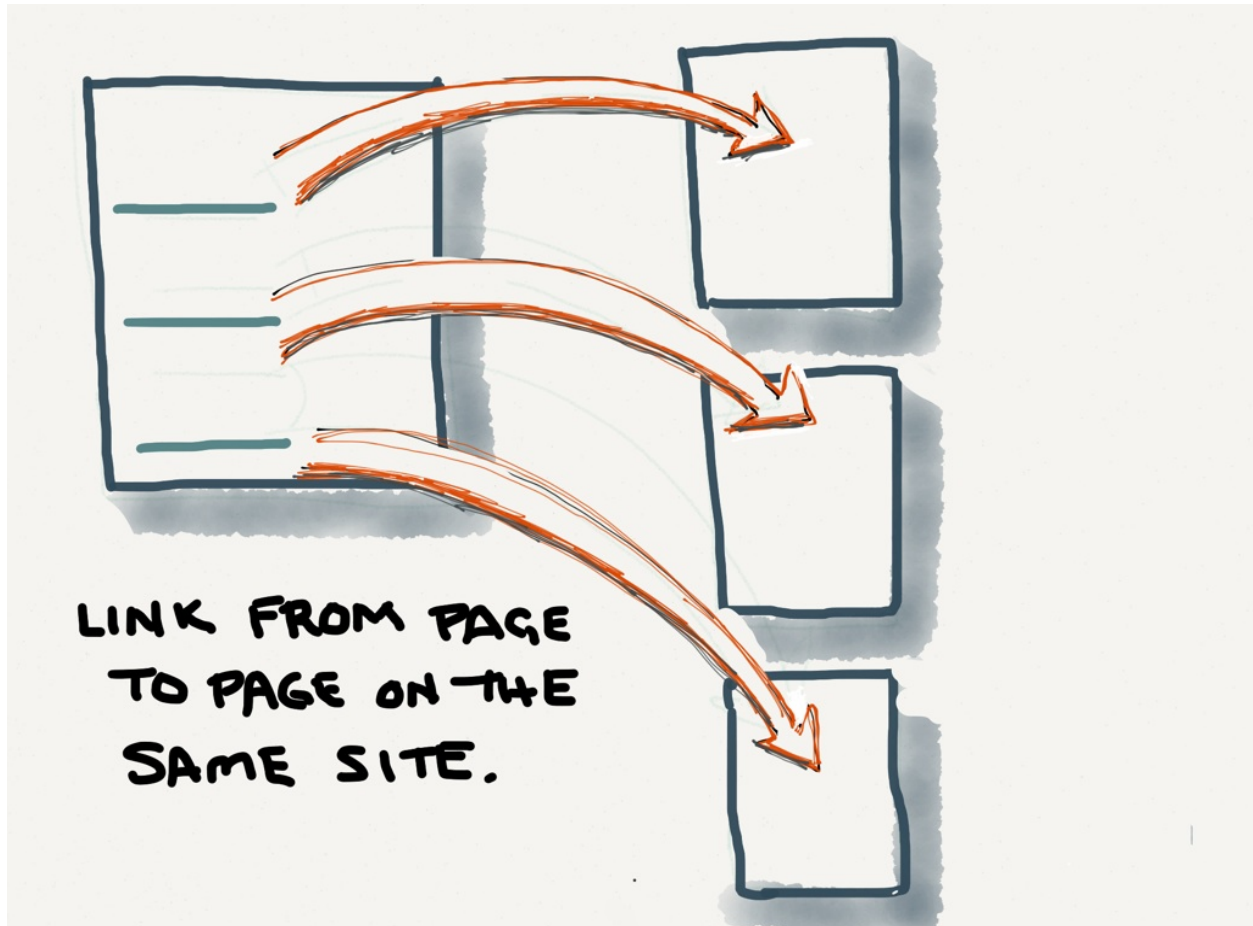
-- Wikipedia http://en.wikipedia.org/wiki/Hyperlink#History_of_the_hyperlink



What You Will Learn

- The basic pattern for every link and anchor.
- Linking to another page on your site.
- Linking to a page out on the Web.
- Setting an anchor.
- Linking to an anchor.
- Linking to an anchor on another page.
- Create links to download files to the user's computer
- Creating an email link.
- Styling links using CSS.
- How to use links with images.

Link to a page on your site.



This is the basic pattern of the anchor or hyperlink element. The `href=" "` attribute tells the browser where to jump to.

```
<a href="contact.html">
```

Contact Us

```
</a>
```

The text in between the start anchor "`<a>`" tag and the stop anchor "`</a>`" is what displays on the page. Usually this is blue underlined text for unvisited sites and purple underlined text for pages that you have already viewed.

However, the appearance of any link can be easily changed using CSS.

In this example the browser will look for a page named `contact.html` located in the same folder as the page that contains the hyperlink.

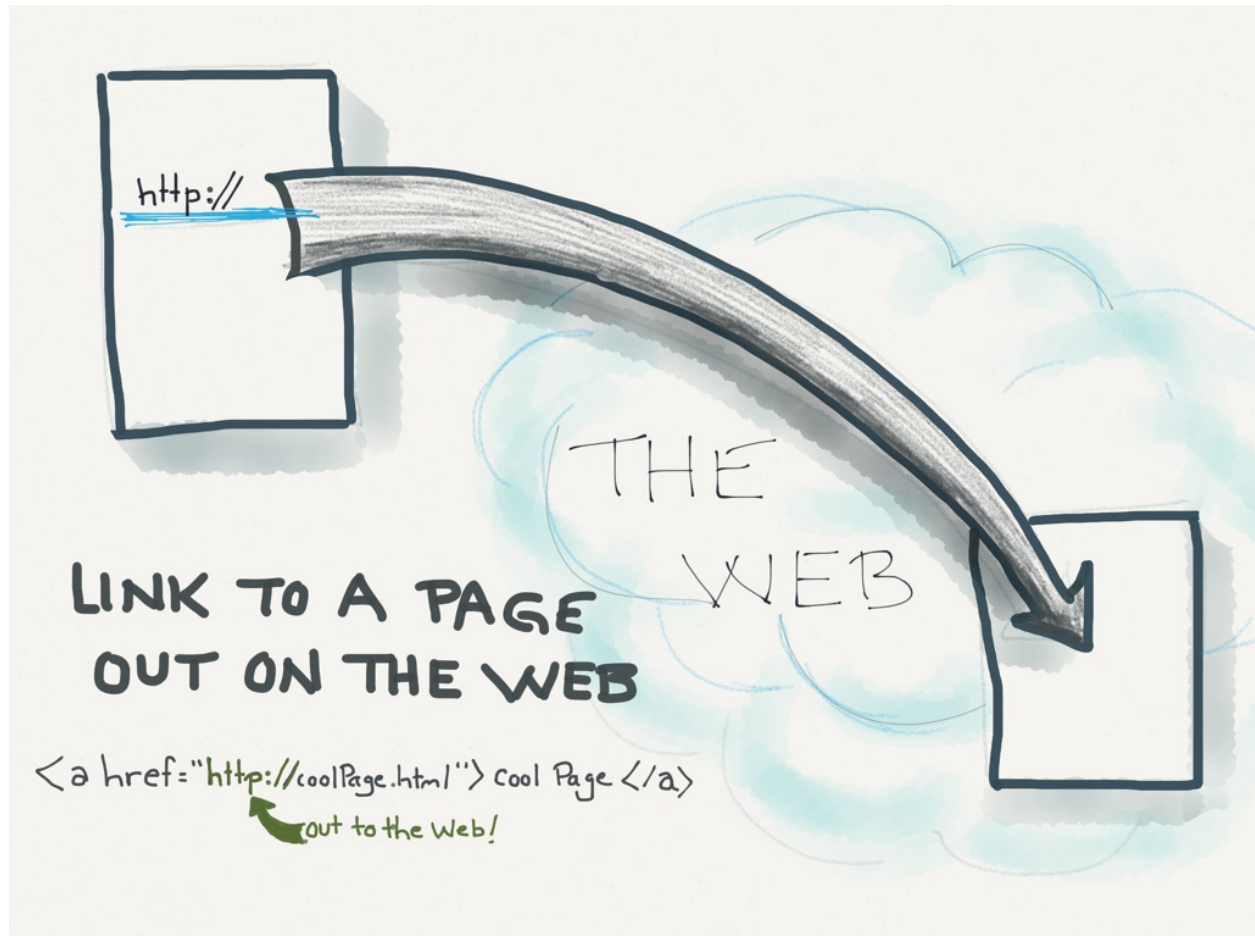
Often the `<a>` element is written on a single line like this:

```
<a href="contact.html">Contact Us</a>
```

Just don't forget the closing bracket `>` for the start tag.

And, if all the sudden the entire page is a hyperlink... then you forgot to type in a matching stop tag `</a>` someplace.

Link to a page out on the Web.



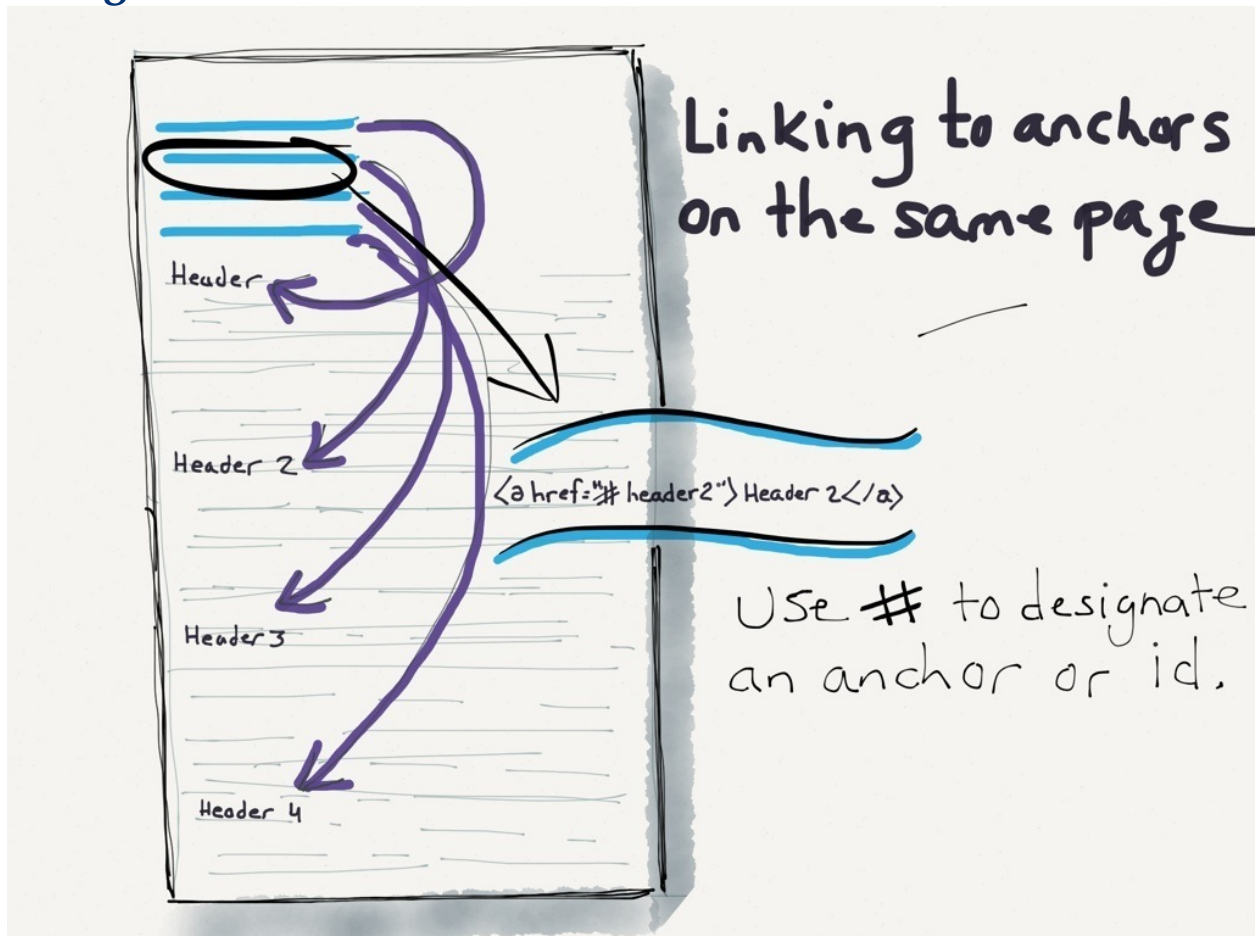
Use the same format to link to a page out on the Web. The only difference is that you add `http://` and the domain name, telling the browser which browser out on the Web should receive the page request.

```
<a href="http://anotherWebSite.com/
contact.html">
    Contact Us
</a>
```

http:// is a protocol that tells the browser to send a request to the server another [anotherWebSite.com](#) at port 80 for the file named contact.html that will be located in the root directory of the site.

If the site is a secure site then the domain name would use https:// and the request would go to port 443 on that web server.

Using anchors.



If you want to set up links to different parts of a web page you would set up anchors and then display hyperlinks to those anchors.

This is common when a page has lots of text such as Wikipedia articles.

There are two ways to set an anchor.

You can use the anchor element `<a>` using the `name=""` attribute instead of the `href=""` like this:

```
<a name="topOfPage"></a>
```

If you want the anchor to be invisible, simply put the stop anchor `</a>` tag

right after the start tag.

You can set an `id=""` attribute inside any other element

```
<h1 id="introducing">  
    Introducing the Greatest Widget Since  
the Printing Press  
</h1>
```

When you use the `id=""` attribute you have to use a unique name. No other attribute can use this name. Also, UPPER/lower case matter so follow a consistent naming style. You can see that the same text from the heading is used to create a unique id for the `h1` tag.

To jump to any anchor simply add the id marker `"#"` to the name:

You might put this at the bottom of the page so users could quickly jump back to the top:

```
<a href="#topOfPage">  
    Jump to the top of the page.  
</a>
```

Or you could use this to jump the user back to that particular `<h1>` element on the page:

```
<a href="#introducing">
```


[Return to Introduction](#)

Link to an anchor on another page.

Many times you want to jump to a specific anchor on another page. This commonly happens when there is a table of contents for the site or a menu item.

For example, the user is on the home page and sees a menu item to the customer service department which is located on the contact page.

The link would include the name of the contact.html page as well as the anchor (#customerService) located on that page to the customer service department.

It might look something like this:

```
<a href="contact.html#customerService">  
    Contact Customer Service  
</a>
```

This link will first jump to the contact.html page and then jump down to the anchor or id named customerService.

Links to Download Files

When a file extension is unknown to the browser it will automatically ask the user if he or she wants to download the file.

Examples of this would be Excel files (.xlsx), Word documents (.docx), PowerPoint files (.pptx) and sometimes Adobe PDF files. Most browsers are not written to handle these particular file formats so the next best alternative is to download it to the user's computer so he or she can run the program needed for that file format.

UX (User eXperience) - It is always helpful if the user knows that the link connects to a specific file type. You can do this by (1) including the actual file name and (2) adding a small icon depicting the file type. Simply add the `<img />` inside the hyperlink.

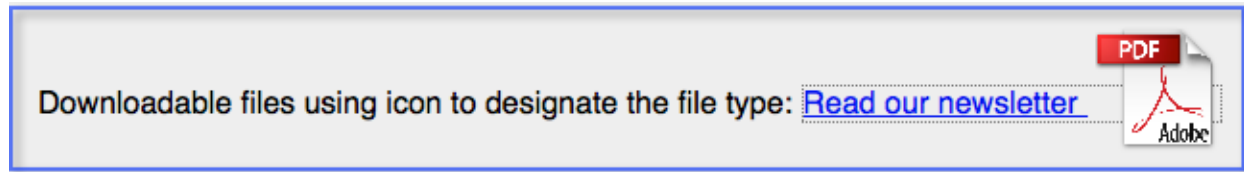
You can get icons at [Komodo's site](#) as well as at [famFamFam.com](#).

The code might look something like this:

```
<a href="newsletter.pdf">  
  Read our newsletter  
      
</a>
```

Notice that that inline CSS was used with the `style=" "` attribute. The `-20px` moves the image down so it centers with the text.

Here is what the link will look like:



Create an email link

You can also set up a link that will automatically open an email program using the mailto: protocol.

```
<a href="mailto:marissa.mayer@yahoo.com">  
    Send an email to Marissa  
</a>
```

When the browser sees this code it will try to run up the resident email program, creating a new email to send out.

In today's world of online email such as gmail.com and yahoo.com this often fails because there is not a local email program. Depending on which browser the person is using the online service may start up but this is not always guaranteed.

Also, software programs known as spiders roam from link to link across the Web looking for the mailto: text as well as the @. Once they find it they can easily extract out the email address and send it to a spam database.

A nice way to obfuscate (confuse) the spam spiders is by using CSS:

Here is a typical email example: missySippy@longRiver.com The spam spiders can see the email in both the text on the page as well as in the href=" " attribute!

To hide the email first set the CSS like this in the <style> element at the top of the page (inside the <head> element):

```
a span.displayNone {
display:      none;
}
```

Then, where you display the email hyperlink insert a null (shown in brown) between the first half of the email and the domain name portion. Use single quotes for the for the class=' ' attribute inside the :

```
<a href="mailto:missySippy@<span class='displayNone'>null</span>longRiver.com">
    Contact Missy Sippy via email.
</a>
```

You could also use the special character @ (shown in brown) to replace the @ sign although most spam spiders are smart enough to catch that.

```
<a href="mailto:missySippy&#64;<span
class='displayNone'>null</span>longRiver.com">
    Contact Missy Sippy via email.
</a>
```

Note that the element uses single quotes so they don't short-circuit the double quotes already in play with the href=" " attribute.

Also, don't use your email as the actual text that shows up on the page. Spam spiders can read any text on the page and displaying your email on the page itself defeats the purpose of fooling the spiders. Instead use a phrase such as: Contact Missy Sippy via email.

Here are more details.

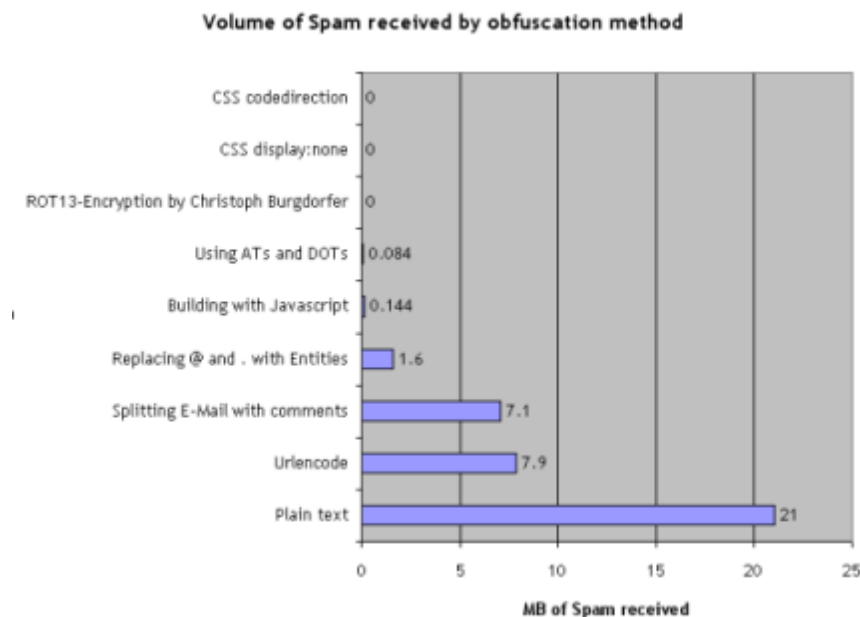
In 2006 Silvan Muhlemann opened nine different e-mail addresses. He published nine e-mail addresses online. Each one was obfuscated by a

different method. He made sure the page was getting indexed by Google.

Then he waited 1.5 years.

For each e-mail address he counted the amount of spam he received. The amount of spam received started by 21MB (for no obfuscation and a total of over 1800 spam mails) and went down to absolutely no spam.

Here is a graph showing the results. The "CSS display none" method is second from the top. You can see the other methods he used by reading the original article: <http://techblog.tilllate.com/2008/07/20/ten-methods-to-obfuscate-e-mail-addresses-compared/>



Open Link in New Page target=" "

Often you don't want to lose the customer to the hinterlands of the Web never to return.

In the <a> tag include the target="_blank" attribute (highlighted in brown below) to automatically open the page in a new page or tab.

```
<a href="http://apple.com"  
target="_blank">Visit the Apple website</a>
```

User's can do this by right-mouse clicking on a link and choosing "Open in new tab" but most users aren't aware of this option.

The downside of this technique is that a LOT of windows/tabs get opened which can be frustrating and confusing to a user who is not comfortable closing a window when she or he is done with it.

The browser tabs show: "The Definitive Guide To Styling", "Super Awesome Buttons with CSS", "Hyperlinks Using CSS", and "Hyperlinkstrukturen.png". The address bar shows the file path: `rj1/Desktop/macExternal/courseContent/classWeb4Biz/09hyperlink/hyperlinkUsingCSS.html`.

Hyperlinks Using CSS

Plain hyperlink: [Read my Blog](#) - Border added to links to show the size of the "hot" area.

Remove underline + add padding for larger target area: [Read my Blog](#)

Change link color: (all four states: new - visited - hover - active) [Read my Blog](#)

-- Depends on browser implementation, especially for visited links

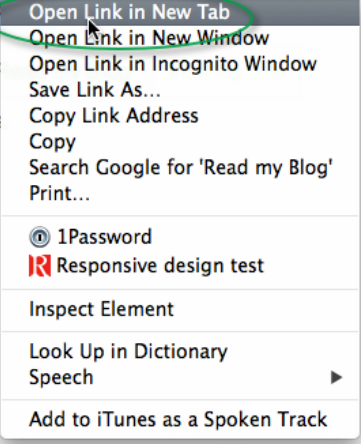
Button Effect: [Read my Blog](#)

Hover Effect: [Read my Blog](#)

Button Effect using a graphic with text: 

Whatever effect you use, be consistent and use the same effect for all the hyperlinks on the page.

Right-mouse click



Styling Hyperlinks with CSS

Joining CSS with hyperlinks creates a whole new look and feel to your web pages.

One of the most common examples are CSS-based menus. They are simply hyperlinks in an unordered list, styled using CSS.

Smashing Magazine offers several things you can do with hyperlinks to enhance UX (User eXperience)

<http://www.smashingmagazine.com/2010/02/13/the-definitive-guide-to-styling-web-links/>

Here is a standard, plain-vanilla link you may have on your site:

```
<a href="http://yourOwnDomainName/blog">
    Read my Blog
</a>
```

Add popup descriptions by adding the title attribute to the hyperlink.

```
<a href="http://yourOwnDomainName/blog" title="Click
here to read my blog">
    Read my Blog
</a>
```

Here is the CSS code that will enhance the UX of your links. You will probably want to add a class=" " or id=" " attribute to individual links so you can focus the CSS on just those items on the page.

Keep the appearance consistent by including each of these in your CSS.

Remove the underline

```
a {
text-decoration: none;
```

```
}
```

Make the link area larger

This is especially useful for mobile devices. The added CSS is highlighted in brown.

```
a {
text-decoration:    none;
padding:           10px;
}
```

Change the color of a link

Use a lighter color for new links and a darker shade for visited links. Make certain there is contrast so the hyperlinks stand out on the page, different from the rest of the text.

There are several states a link can be in:

- new
- visited
- hover over
- active.

These are more specific settings so make sure they come after the more general CSS code that is used to style all four states.

```
a:link,
a:hover {
color:           #4EC044;      /* lighter green */
}

a:visited,
a:active {
color:           #008000;      /* forest green */
}
```

Make a link look like a button

First, add a class to the links you want to "buttonize". Using a class means you can use that attribute on several buttons on the web page.

```
<a href="http://yourOwnDomainName/blog"
class="buttonize" title="Click here to read my blog">
    Read my Blog
</a>
```

```
.buttonize{
display: inline-block;
padding: 5px 10px 6px;
color: #fff;
text-decoration: none;    /* Remove underline */
font-weight: bold;
line-height: 1;

/* Round the corners for various browsers */
-moz-border-radius: 5px;
-webkit-border-radius: 5px;

/* Add a drop shadow for various browsers
   User rgba( ) to automatically convert
   depending on light or dark background */
-webkit-box-shadow: 0 1px 3px #999;
-moz-box-shadow: 0 1px 3px rgba(0,0,0,0.5);
-webkit-box-shadow: 0 1px 3px rgba(0,0,0,0.5);
text-shadow: 0 -1px 1px rgba(0,0,0,0.25);

border-bottom: 1px solid rgba(0,0,0,0.25);
position: relative;
cursor: pointer;
}
```

For more details check out the Zurb blog: <http://zurb.com/article/266/super-awesome-buttons-with-css3-and-rgba>

Give the user immediate feedback for each click

Use the active: state to provide visual feedback to the user to indicate that they have clicked a link. As they are waiting for the new page to display they will have visual confirmation that "Yes, you did click the link. Your page is coming soon!" One nice effect is to move the link down one or two pixels, giving the appearance of being pressed.

Again, this is more specific, only to the :active link, so make sure this CSS code is below the more general rules that are used to style all the states.

```
a.active {  
padding-top:      2px;    /* move the link down to show  
it has been clicked */  
}
```

This may not work on all browsers. Test different browsers to be sure.

Use the hover effect to give feedback to the user

as they mouseover a link. (Not very useful on most mobile devices)

```
a:hover {  
text-shadow: 0 0 2px #999;  
}
```

Here is a sample web page showing how these techniques can be used. This is a sampler showing lots of possibilities. You'll notice how there is a slight confusion as you work with the buttons, wondering what effect is going to happen.

When you code your pages, set up one set of styles and use them consistently with every hyperlink on the page.

Hyperlinks Using CSS

Plain hyperlink: [Read my Blog](#) - Border added to links to show the size of the "hot" area.

Remove underline + add padding for larger target area: [Read my Blog](#)

Change link color: (all four states: new - visited - hover - active) [Read my Blog](#) [Visit my Site](#)

-- Depends on browser implementation, especially for visited links.

Button Effect: [Read my Blog](#)

Hover Effect: [Read my Blog](#)

Button Effect using a graphic with text: [Like This](#)

Whatever effect you use, be consistent and use the same effect for all the hyperlinks on the page.

Here is the code for these examples:

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <!-- hyperlinkUsingCSS.html - Demonstration showing how
anchors can be enhanced using CSS
  Peter Johnson - peterk@webExplorations.com
  Written:    08-08-13
  Revised:    10-27-13 PKJ Set up unique id for each
type of button
  -->
  <title>Hyperlinks Using CSS</title>

  <style type="text/css">
```

```

/* plain hyperlink - add border for visual reference
*/
a {
border:                1px #777 dotted;
}

/* hyperlink without the underline */
#noUnderline:link,
#noUnderline:visited,
#noUnderline:hover,
#noUnderline:active{
text-decoration:      none;
padding:              10px;
}

/* change link color based on state */
.colorChange:link {
color:                #4EC044;      /* lighter green */
text-decoration:      none;
padding:              10px;
}

.colorChange:visited,
.colorChange:active {
color:                #008000;      /* forest green */
text-decoration:      none;
padding:              10px;
}

#like {
text-decoration:      none;
color:                #6983AA;      /* steel blue */
padding:              10px;
}

#like:visited {
color:                #008000;      /* forest green */
}

```

```

#like:hover {
color:                black;
}

.buttonize{
display:              inline-block;
padding:              10px 15px;
color:                #EEE;
background:           #008000;
text-decoration:      none;    /* Remove underline */
font-weight:          bold;
line-height:          1;

    /* Round the corners for various browsers */
-moz-border-radius: 5px;
-webkit-border-radius:5px;
/* add shadow effect to box */
-moz-box-shadow:      0 1px 3px #999;
-webkit-box-shadow:   0 1px 3px #999;
border-bottom:        1px solid #222;
position:              relative;
cursor:                pointer;
}

.buttonize:hover {
color:                #D1DFF5;    /* light gray blue */
text-shadow:          0 0 5px #000080;
}

#hovering:hover {
text-shadow:          0 0 2px #999;
}

body {
background:           #EEE;

```



```

font-size:          16px;
font-family:        helvetica, arial, sans-serif;
}

h1 {
border:              1px #000080 solid;
padding:             10px;
background:          #AAA;
color:               white;
}

p {
margin:              2em 0px 0px 0px;
}

#frame {
max-width:           800px;
margin:              auto;
border:              1px solid #000080;
padding:             20px;
}
</style>
</head>
<body>
<div id="frame">
<h1>Hyperlinks Using CSS</h1>
<!-- actual href removed to prevent jumping to another
page -->
<p>
  Plain hyperlink: <a href="http://webExplorations.com/
blog" title="Read my blog">Read my Blog</a> - Border
added to links to show the size of the "hot" area.
</p>
<p>
  Remove underline + add padding for larger target
area: <a href="http://webExplorations.com/blog"
title="Read my blog" id="noUnderline">Read my Blog</a>
</p>

```

```
<p>
  Change link color: (all four states: new - visited -
  hover - active) <a href="http://webExplorations.com/
  blog" title="Read my blog" class="colorChange">Read my
  Blog</a>
  <a href="http://webExplorations.com"
  class="colorChange">Visit my Site</a>
  <br /><br />
  -- Depends on browser implementation, especially for
  visited links.
</p>
```

```
<p>
  Button Effect: <a href="http://webExplorations.com/
  blog" title="Read my blog" class="buttonize">Read my
  Blog</a>
</p>
<p>
```

```
  Hover Effect: <a href="http://webExplorations.com/
  blog" id="hovering">Read my Blog</a>
</p>
```

```
<p>
  Adding a graphic with text: <a href="http://
  webExplorations.com/blog" id="like">
    Like This
  </a>
</p>
```

```
<p>
  Downloadable files using icon to designate the file
  type:
  <a href="newsletter.pdf">
    Read our newsletter
    
```

```
</a>
</p>
```

```
<p>
<strong>Whatever effect you use, be consistent and use
the same effect for all the hyperlinks on the page.</
strong>
```

```
</p>
```

```
<hr />
```

```
<p>
```

```
Thumbnail images as links to a larger photo:<br />
```

```
<a href="graphic/structure.png" target="_blank">
```

```

```

```
</a>
```

```
<br />
```

```
Please click on the image for a larger view.
```

```
</p>
```

```
</div> <!-- end of #frame -->
```

```
</body>
```

```
</html>
```

Links and Images

Another useful technique is to combine the element as a hyperlink.

For example, you may have a series of photos that you want to display but don't have room for them all on the page. Make a set of thumbnail images and then put a hyperlink around each one. When the user clicks on one of the thumbnails a larger will display.

Make sure you tell the user this option is available. The phrase "Click on the image for a larger view" will let the user know that they no longer have to squint at the images on the page.

Here is what the code will look like.

```
<a href="graphic/myImage.png"  
target="_blank">  
      
</a>  
<br />  
<p>  
Please click on the image for a larger  
view.  
</p>
```

The inside image is a small thumbnail version of the original image.

Both images are stored in the sub-folder named "graphic"

When the user clicks on the link the larger image is displayed on a new page or browser tab.

Best Practices

Follow these guidelines:

Absolute vs Relative

When writing a hyperlink, don't refer to files using an absolute filenames. Use relative names instead.

For example, when linking to a file on the same web site:

```
<a href="C:\contact.html">Contact Us</a>
```

will not work when the file is moved up to the web server because there is no C:\ drive out on the server.

Instead use relative naming:

```
<a href="contact.html">Contact Us</a>
```

This tells the browser to look in the same folder for contact.html as the web page containing the hyperlink.

If the file was in a sub-folder, such as customerService, then that folder has to be named:

```
<a href="customerService/contact.html">Contact Us</a>
```

Keep your site self-contained.

All the links should be within the same folder structure. This allows you to move a folder to another server or another computer without breaking a lot of links and images.

Use descriptive language for each hyperlink to improve SEO.

[Click Here](#) doesn't tell the user (or the search engines checking your SEO) what the link is about. Use a descriptive phrase letting the user know where the link will be displaying. For example, [View this spreadsheet showing the current employment surge in Web Programming](#).

Only use underlining for hyperlinks.

When a person sees underlined text they automatically assume that it is a link. As soon as he or she clicks on it they discover that it is only underlined text and not a hyperlink, they feel stupid. Don't fool people with underlining. Use bold and/or italics instead, especially for book titles.

Summary

What have you learned?

- How to link to a file on the same site.
- How to link to a file out on the web.
- Setting up anchors using the <a> element as well as the id=" " attribute.
- How to link to an anchor.
- How to link to an anchor on a different page.
- How to create an email link and fool the spam spiders at the same time.
- Create links to download files to the user's computer
- How to make a link look like a button and other CSS styling tricks.
- How to link images.
- Some best practices.

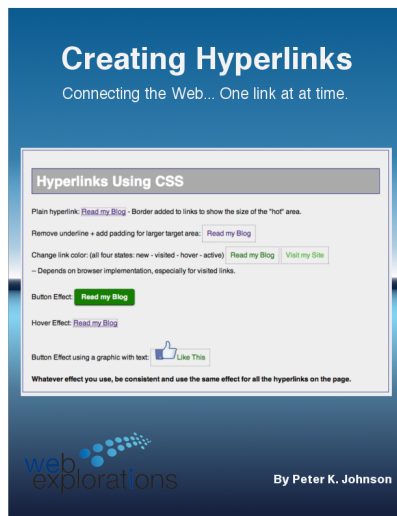
Credits

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Peter K. Johnson - <http://WebExplorations.com>

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Revised: November 18, 2013 - updated spam email link.



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